

# **Canadian Households Housing Burden**

*A Business Consultancy Report*

Atom Li

## **Table of Contents**

1. Executive Summary
2. Introduction and Problem Statement
3. Data Preparation
4. Unsupervised Learning
  - 4.1 Exploratory Data Analysis
  - 4.2 K-Means Clustering
  - 4.3 PCA
  - 4.4 UMAP
5. Supervised Learning
  - 5.1 ElasticNet GLM
  - 5.2 XGBoost
  - 5.3 SHAP Analysis
6. Recommendations and Conclusions

## **1. Executive Summary**

This report examines the Housing Burden Ratio (HBR) across Canadian dissemination areas using Environics Analytics data. After merging household spending and demographics, filtering to the PRCDDA level yields 57,936 areas with 3,432 features. Unsupervised learning identifies two segments: Cluster 0 (44,540 areas, economically distinct) and Cluster 1 (945 areas, average households). XGBoost achieves a test  $R^2$  of 0.7411, outperforming ElasticNet ( $R^2 = 0.4365$ ). SHAP analysis reveals the relationship is fundamentally non-linear, driven by income and expenditure variables.

## **2. Introduction and Problem Statement**

Housing affordability is a pressing issue for Canadian households. When too much disposable income goes toward shelter, the capacity for other needs shrinks. This report asks: what factors determine housing burden, and can we predict it? We use data preparation with leakage removal, unsupervised learning (K-Means, PCA, UMAP), and supervised models (ElasticNet, XGBoost with SHAP). The data is cross-sectional, so our analysis is diagnostic rather than fully predictive.

## **3. Data Preparation**

The raw data consists of Household Spending (two CSVs, 268 and 182 columns) and Demographic Statistics (six CSVs), each with ~950,000 rows. We filtered to PRCDDA (Dissemination Area) using Polars LazyFrames to avoid loading full files into RAM. The merged dataset contains 57,936 rows and 3,432 columns.

$HBR = HSSH001S / HSAGDISPIN$ . To avoid leakage, we removed all HSSH columns and seven aggregates (HSAGDISPIN, HSAGDISCIN, HSHNIAGG, HSTT001, HSTE001, HSTX001, HSTC001). After removing CODE and GEO and invalid rows, the final tableau has 56,857 rows and 3,429 features. No zero-variance columns or missing values were found.

## **4. Unsupervised Learning**

### **4.1 Exploratory Data Analysis**

After standardization using StandardScaler, we split the data 80/20 into training (45,485 rows) and test (11,372 rows) sets with `random_state=42`. An 80/20 split provides sufficient training data while reserving a large enough test set for reliable evaluation. Median imputation was chosen over mean because household expenditure variables tend to be right-skewed, making the median more robust to outliers and a better representation of the typical value. Zero-variance columns carry no discriminative information and were checked for removal, though none were found in this dataset.

The target variable HBR has a skewness of 15.407, indicating an extremely right-skewed distribution. Most dissemination areas have a moderate housing burden (concentrated around 0.1–0.3), but a long tail extends past 4.0. This skewness has implications for model choice, as linear models struggle with such distributions.

Correlation analysis on the training set reveals that the top predictors of HBR are low-income household counts (ECYHRI\_020, P3YHRI\_020, P0YHRI\_020, P5YHRI\_020 — all representing households with income \$0–19,999), vehicle purchase proxies (HSRV003B for motorcycles/snowmobiles, HSRV004 for ATVs), and current-year income measures (ECYHNI\_020). Low-income bracket variables dominate, suggesting housing burden is most strongly associated with concentrations of very low-income households.

## 4.2 K-Means Clustering

We ran K-Means for  $k=2$  through 12. Silhouette scores were computed on a 5,000-row random sample because the silhouette metric computes pairwise distances between all points, which is very slow on the full 45,485 rows. This sampling does not change the ranking of  $k$  values, just makes the scores slightly approximate. K-Means itself was still fit on all training data. The elbow curve shows a sharp drop from  $k=2$  to  $k=3$ , with diminishing returns afterward. The silhouette score peaks at  $k=2$  (0.7472), dropping to 0.5128 at  $k=3$  and 0.3399 at  $k=4$ . Both methods agree that  $k=2$  is optimal. When both point to the same  $k$ , we have stronger confidence. If they had disagreed, we would have favoured the silhouette score since it directly measures cluster quality.

Our preprocessing choices can affect the optimal  $k$ . Since we standardized all features, each one has equal weight in the distance calculations. If we had applied log transforms to skewed features or used a subset of features, the distances would change, potentially shifting the optimal  $k$ . With 3,429 features, the high dimensionality may dilute cluster structure, which tends to favour a smaller  $k$ .

The two clusters are highly imbalanced: Cluster 0 contains 44,540 observations and Cluster 1 contains 945. This indicates the data is mostly homogeneous, with Cluster 0 being around 47 times the size of Cluster 1. We selected the top 7 features most correlated with HBR: low-income household counts (ECYHRI\_020, P3YHRI\_020, P0YHRI\_020, P5YHRI\_020, ECYHNI\_020) and recreation spending proxies (HSRV003B, HSRV004). HSRV003B and HSRV004 are proxies of above-average lifestyle spending, while the income variables are proxies of below-average income, as they represent the lowest income group. Low income has a strong correlation with HBR because lower-income individuals may have a harder time paying for their household. On the other hand, spending on unnecessary items like motorcycles and ATVs also has a strong correlation with HBR, as they are indicators of high spending that the average person is not in a position to partake in.

Cluster 1 shows standardized feature means around 2.0–2.5 across all seven variables, versus near-zero for Cluster 0. Cluster 1 (tight distribution) is representative of average households, while Cluster 0 (big spread, extreme HBR) represents households with higher values in the variables most correlated with HBR — the edge cases. We name Cluster 0 the “economically distinct household cluster” and Cluster 1 the “average household cluster.” The economically distinct cluster, being 47 times the size of the average cluster, represents a subgroup of extreme households distinct from typical household values.

A boxplot of HBR by cluster shows that while both clusters have similar median housing burden, Cluster 0 exhibits far greater variability and more extreme outliers. This indicates more inequality within the larger cluster. The economically distinct households in Cluster 0 capture a much more spread, with many outlier areas experiencing extreme housing burden above the mean. This makes sense because these areas have higher values in the variables most correlated with HBR. On the other hand, Cluster 1 (average households) shows less variability with fewer outlier areas experiencing financial strain. This cluster contains a very specific type of household grouped by K-Means due to similarity in feature space, resulting in a tight distribution and low variability. We can conclude that the clusters capture meaningful differences in terms of edge cases and variability, even though the median values are close.

### 4.3 PCA

The top 10 principal components explain 65.6% of the total variance. This is moderate for 3,429 features, suggesting the data is genuinely high-dimensional.

Are the clusters clearly separated? They are not. It looks like one big cluster with a line somewhat dividing it up, and each of the two divisions is represented by a different colour. On either side of the division, there is a slight overlap from the other cluster, where we place the imaginary division line. The data does not satisfy the cluster assumption due to overlap. There are no distinctly separated groups, as it appears as one cluster with left and right-sided subgroups.

For the top 3 PCs: PC1 loads on P3YEDUHP15, P3YCWHP1, P3YBAS15HP, P3YMARHP15, and P3YBAS12HP. This represents overall household population composition based on age 12+/15+, including educational attainment, marital status, and class of worker. PC2 loads on P5YHTAMED, EGYPTAMED, P3YPTAMED, P3YHTAMED, and EGYHTAMED — all median age variables. This represents the age profile of households, capturing variation in median age. PC3 loads on EGYHRIMED, EHYHNI2040, EHYHRI2040, EHYHRIMED, and EHYHNIMED. This represents household income levels, particularly differences in the \$20,000–39,999 range.

Mean PC scores by cluster show Cluster 1 has higher PC1 (0.587 vs 0.123), lower PC2 (−0.546 vs 0.001), and lower PC3 (−0.620 vs −0.003). Based on these PCA concepts, the proposed

human-readable cluster names are: Cluster 0 as “Economically and demographically distinct households (older, higher-income)” and Cluster 1 as “Average households (younger, lower-income, typical composition).”

How do these differ from the previous ones? The cluster names based on the PCA concepts differ from the previous ones for several reasons. First, the new names are human-readable. This allows for better analysis since it saves time for anyone interpreting the code. Without these readable names, someone would need to read through and fully understand what each variable represents. Given how complicated this dataset is, with many components and factors to consider, it would take a lot of effort. They are also more comprehensive as they combine multiple underlying features into broader concepts such as demographic structure, age profile, and income level, rather than relying on individual variables. This allows for a more meaningful and clear interpretation of our clusters.

## 4.4 UMAP

To search for the optimal value of `n_neighbors`, we tested three values (15, 30, 50) and selected the one yielding the highest silhouette score in 2D. The best value was `n_neighbors=50` (silhouette=0.2400), marginally outperforming 30 (0.2342) and 15 (0.2279). We selected `n_neighbors=50` to balance local and global structure on this large dataset. `n_components=2`, `metric='euclidean'`, and `n_epochs=200` are standard for numerical data. `min_dist=0.3` balances detail and spread. `low_memory=True` and `n_jobs=1` were required due to hardware constraints.

Is UMAP a better or worse representation than PCA? UMAP is a better representation than PCA for visualizing the cluster structure in this dataset. In the PCA scatter plot, the two clusters are not clearly separated. Cluster 0 appears as a tight mass on the left, while cluster 1 spreads across a very wide range, with extreme outliers dominating PC1 and making the plot difficult to interpret. There is significant overlap between the two groups and no clear boundary.

In the UMAP plot, cluster 1 (orange) is visibly separated in the bottom-right corner, forming a distinct region away from the large blue mass of cluster 0. This separation was not visible in PCA, which suggests there is a non-linear structure in the data that PCA's linear projection cannot capture, but UMAP's non-linear approach can. Overall, UMAP provides a more interpretable and meaningful 2D representation of the cluster structure.

The silhouette score in 2D UMAP space is 0.2624, compared to 0.7472 in the original high-dimensional space. This drop is expected — compressing thousands of features to two dimensions inevitably loses cluster information. However, UMAP still achieves a positive score and visually separates cluster 1 from cluster 0, suggesting it has preserved the most important structure. The gap tells us the full cluster structure is richer than any 2D projection can represent.

Across all three representations — KMeans labels, PCA, and UMAP — cluster 1 (the smaller, economically distinct group) is consistently the most informative. In K-Means, it was identified as a distinct minority group with extreme values in the variables most correlated with HBR. In PCA, it appears as the widely dispersed group with extreme outliers along PC1. In UMAP, it forms a clearly separated region in the bottom right. The fact that this cluster is identifiable across all three methods, despite their very different approaches, gives us strong confidence that it represents a genuinely distinct subgroup of households rather than an artifact of any single method.

## **5. Supervised Learning**

### **5.1 ElasticNet GLM**

Since HBR is bounded, we transformed the target using a Pearson Type III distribution, mapping values through the PT3 CDF then the inverse normal PPF to produce z-scores (mean  $\approx 0$ , std  $\approx 0.984$ ). Due to computational constraints, we selected the top 500 features by correlation with the transformed target on the training set only, ensuring no leakage.

GridSearchCV (5-fold) searched six alpha values in  $[1e-4, 1.0]$  and five l1\_ratio values in  $[0.1, 0.9]$ , avoiding extremes of 0 and 1 which are slow to converge. Best parameters: alpha=0.0001, l1\_ratio=0.9. Test  $R^2 = 0.4365$ , MSE = 0.002190, bootstrapped 95% CI [0.2204, 0.7072]. The wide confidence interval tells us the model's performance is unstable — it explains some of the variance in housing burden but misses a large portion, likely due to the non-linear relationships that a linear model cannot capture. The predicted vs actual scatterplot shows the model captures the general trend but struggles at higher HBR values where the skewed distribution creates challenges.

The model retains 374 of 500 non-zero coefficients. Top 5 by magnitude: HSRV022 (0.942), Y23NINCAVG (0.802), Y23RINCAVG (0.778), HSRV021 (−0.665), HSRV081 (−0.549). These are reasonable predictors. HSRV022 and HSRV021 relate to recreation vehicle expenditures, while Y23NINCAVG and Y23RINCAVG are projected income averages. The positive coefficients on recreation spending suggest areas with higher discretionary expenditure tend to have higher housing burden, while the negative coefficients on recreation sub-categories suggest specific spending patterns reduce burden. Overall, these tell us that housing burden is driven by the interplay between income levels and spending behaviour.

$R^2$  per cluster: Cluster 0 = 0.4374 (CI [0.2493, 0.7330]), Cluster 1 = −0.0462 (CI [−0.6499, 0.3627]). There are meaningful differences between groups. The model performs reasonably on Cluster 0 but fails on Cluster 1 because it contains average households with less variance to explain. The negative  $R^2$  means predictions are worse than the mean. Given our understanding of these groups, this makes sense: the economically distinct cluster has more variability in HBR for

the model to learn from, while the average cluster's tight distribution gives the linear model little signal to work with.

## 5.2 XGBoost

XGBoost uses HBR directly (no PT3 transform needed). RandomizedSearchCV (5-fold, 20 iterations) searched: `n_estimators` (100–500), `max_depth` (3–6), `learning_rate` (0.01–0.2), `subsample` (0.6–1.0), `colsample_bytree` (0.6–1.0), with `tree_method=hist`. Best: `subsample=0.8`, `n_estimators=500`, `max_depth=4`, `learning_rate=0.1`, `colsample_bytree=0.6`. Test  $R^2 = 0.7411$ , MSE = 0.001006, 95% CI [0.6201, 0.9240].

The XGBoost model substantially outperforms the ElasticNet linear model from Part 3a. This performance gap tells us a few things. Non-linear relationships exist between household characteristics and housing burden. Feature interactions matter — XGBoost's tree-based structure picks up complex interactions that linear models miss. The housing burden problem is inherently non-linear. Threshold effects, like income crossing certain brackets, likely drive changes in spending patterns.

XGBoost handles skewed distributions natively and captures threshold effects where income changes non-linearly impact housing burden. The predicted vs actual plot shows tight clustering around the diagonal with no systematic bias, and residuals show constant variance. XGBoost's CI [0.6201, 0.9240] is both higher and tighter than ElasticNet's, meaning XGBoost is more accurate and consistent.

## 5.3 SHAP Analysis

SHAP values were computed using TreeExplainer. Top 10 features by mean absolute SHAP: HSRV003B (0.0119), HSHE012C (0.0052), HSED006 (0.0043), HSWH040S (0.0043), HSOI040Z (0.0042), ECYPNIAVG (0.0039), HSCS004 (0.0037), Y22NINCAVG (0.0036), HSWH021 (0.0029), HSWH040 (0.0029).

The problem is significantly non-linear. SHAP dependence plots were created for the top two features: HSRV003B and HSHE012C. We selected the top two features: HSRV003B (highest SHAP) and HSHE012C (second-ranked, capturing equipment expenditures). The HSRV003B dependence plot shows a curved relationship where the SHAP value increases steeply at low feature values, then flattens, indicating diminishing marginal impact. The HSHE012C plot shows a similar non-linear pattern, with the effect on housing burden varying depending on the interaction with other features (shown by the colour gradient). These curved patterns confirm that linear models miss important structure in the data. XGBoost outperforms ElasticNet ( $R^2$  0.7411 vs 0.4365) precisely because it captures these non-linear relationships.



Top 5 SHAP features (HSRV003B, HSHE012C, HSED006, HSWH040S, HSOI040Z) differ from top 5 ElasticNet coefficients (HSRV022, Y23NINCAVG, Y23RINCAVG, HSRV021, HSRV081). SHAP captures non-linear effects and interactions that ElasticNet's linear framework misses.

The features driving housing burden prediction (income, expenditures) are not the same as those separating clusters (demographics, age, geography). Predicting housing burden requires direct economic inputs, while segmenting households uses stable demographic traits. Two demographically identical households can have very different housing burdens due to income differences. This confirms housing burden is a distinct economic phenomenon, not simply a function of household type.

## 5.4 Cluster Analysis

Model performance varies across clusters. Cluster 0 shows a higher  $R^2$ , suggesting more predictable housing burden patterns. Cluster 1 shows lower  $R^2$ , indicating more variability or missing predictive features. These differences validate that our unsupervised learning successfully identified household segments with different economic realities.

Are the features driving housing burden the same as those separating clusters? No — they capture complementary aspects. Prediction drivers (SHAP) focus on income and expenditures, while cluster separators (PCA) relate to demographics and geography. Housing burden prediction requires real-time economic data, while household segmentation uses stable demographic traits. This confirms housing burden is not simply a function of household type but a distinct economic phenomenon requiring its own modelling approach.

## 6. Recommendations and Conclusions

Based on our analysis, we offer the following recommendations:

**Target income thresholds.** Housing burden is most strongly associated with areas containing high concentrations of households in the \$0–19,999 income bracket. Policy interventions such as housing subsidies or rent controls should prioritize these communities.

**Recognize non-linearity.** The relationship between household characteristics and housing burden is fundamentally non-linear. XGBoost ( $R^2 = 0.7411$ ) substantially outperforms ElasticNet ( $R^2 = 0.4365$ ). Decision-support tools should use non-linear models rather than linear regression.

**Address within-segment inequality.** Cluster 0 (44,540 areas) shows enormous variability in housing burden. Blanket policies would miss the most vulnerable households within otherwise average-looking areas. Granular, area-level targeting is essential.

**Separate segmentation from prediction.** The features that segment households (demographics, age, geography) differ from those predicting burden (income, expenditures). Effective policy needs both segmentation to identify target populations and predictive models to estimate severity within each segment.

**Use XGBoost with SHAP for transparency.** The XGBoost model provides both accuracy and interpretability through SHAP. We recommend this combination for predicting and understanding household expenditure patterns.

In conclusion, the housing burden in Canada is a complex, non-linear phenomenon driven by income levels and expenditure patterns. Unsupervised methods meaningfully segment households, but the features defining segments differ from those predicting burden. Effective policy must both identify at-risk communities and predict burden severity. The XGBoost model with SHAP provides the tools needed for evidence-based interventions.